

LABORATORY OBSERVATION

Course	Full Stack Web Development
Course Code	23CM4121
Observation Task No.	4
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1. TITLE

DOM Manipulation Using JavaScript

2. AIM

To demonstrate how JavaScript uses the Document Object Model (DOM) to dynamically change the content, style and attributes of HTML elements.

3. OBJECTIVE

1. To understand the Document Object Model.
2. To access HTML elements using JavaScript.
3. To change HTML content dynamically.
4. To modify CSS styles using JavaScript.
5. To change HTML attributes using `setAttribute()`.
6. To use functions and event handlers for interactive DOM manipulation.

4. THEORY

The Document Object Model represents an HTML document as a tree of objects. JavaScript can access these objects and modify their content, styles and attributes. `getElementById()` selects an element, `textContent` changes its content, style properties modify presentation, and `setAttribute()` changes an HTML attribute.

Application Flow: HTML Document → DOM Tree → JavaScript Selects Element → Modify Content / Style / Attribute → Updated Webpage

5. ALGORITHM / PROCEDURE

1. Create an HTML document with a heading, paragraph, buttons and image.
2. Create buttons for changing content, style and attributes.
3. Use `getElementById()` to access the required elements.
4. Use `textContent` to replace the paragraph text.
5. Use style properties to modify color, background, size and weight.
6. Use `setAttribute()` to change image attributes.
7. Open the page in a browser and click each button.
8. Verify that the webpage changes dynamically.

6. PROGRAM

```
1: <!DOCTYPE html>
2: <html lang="en">
3: <head>
4: <meta charset="UTF-8">
5: <meta name="viewport" content="width=device-width, initial-scale=1.0">
6:
7: <title>DOM Demonstration</title>
8:
9: <style>
10: body {
11: font-family: Arial, sans-serif;
12: text-align: center;
13: padding: 50px;
14: background-color: #f2f2f2;
15: }
16:
17: .container {
18: background-color: white;
19: width: 500px;
20: margin: auto;
21: padding: 30px;
22: border-radius: 10px;
23: }
24:
25: #message {
26: font-size: 24px;
27: margin: 25px;
28: }
29:
30: button {
31: padding: 10px 15px;
32: margin: 5px;
33: border: none;
34: border-radius: 5px;
35: cursor: pointer;
36: }
37: </style>
38: </head>
39:
40: <body>
41: <div class="container">
42: <h1 id="heading">DOM Demonstration</h1>
43:
44: <p id="message">
45: This is the original content.
46: </p>
47:
48: <button onclick="changeContent()">Change Content</button>
49: <button onclick="changeStyle()">Change Style</button>
50: <button onclick="changeAttribute()">Change Attribute</button>
51:
52: <br><br>
53:
54: 
58: </div>
```

6. PROGRAM (continued)

```
59:
60: <script>
61: // Change the content of the paragraph.
62: function changeContent() {
63: document.getElementById("message").textContent =
64: "The content has been changed using JavaScript!";
65: }
66:
67: // Change the style of the paragraph.
68: function changeStyle() {
69: const message = document.getElementById("message");
70:
71: message.style.color = "blue";
72: message.style.backgroundColor = "lightyellow";
73: message.style.fontSize = "30px";
74: message.style.fontWeight = "bold";
75: }
76:
77: // Change HTML attributes dynamically.
78: function changeAttribute() {
79: const image = document.getElementById("studentImage");
80:
81: image.setAttribute("alt", "Image changed using DOM");
82: image.setAttribute(
83: "title",
84: "This image attribute was changed using JavaScript"
85: );
86: image.setAttribute("width", "300");
87: }
88: </script>
89: </body>
90: </html>
```

7. CODE EXPLANATION

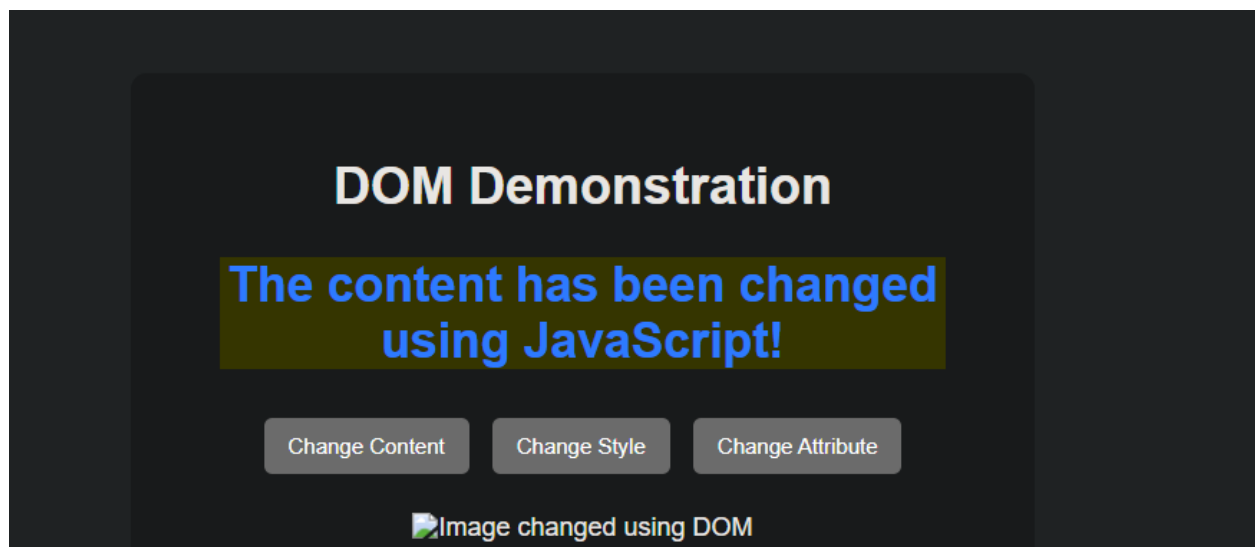
Code / Syntax	Explanation
getElementById()	Selects an HTML element using its unique ID.
textContent	Changes the text content of an HTML element.
style.color	Changes the text color through JavaScript.
style.fontSize	Changes the font size dynamically.
setAttribute()	Adds or changes an HTML attribute.
onclick	Calls a JavaScript function when the button is clicked.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Click Change Content	Paragraph text changes	Content changed	Pass
TC-02	Click Change Style	Paragraph style changes	Style changed	Pass
TC-03	Click Change Attribute	Image attributes change	Attribute changed	Pass
TC-04	Reload and repeat	Page loads without error	Page loads successfully	Pass

9. OUTPUT

DOM Demonstration output after changing the content and style using JavaScript; the Change Attribute action is also available.



10. RESULT

Thus, DOM manipulation was successfully demonstrated using JavaScript. The application dynamically changes HTML content, CSS styling and HTML attributes through JavaScript functions and user interaction.